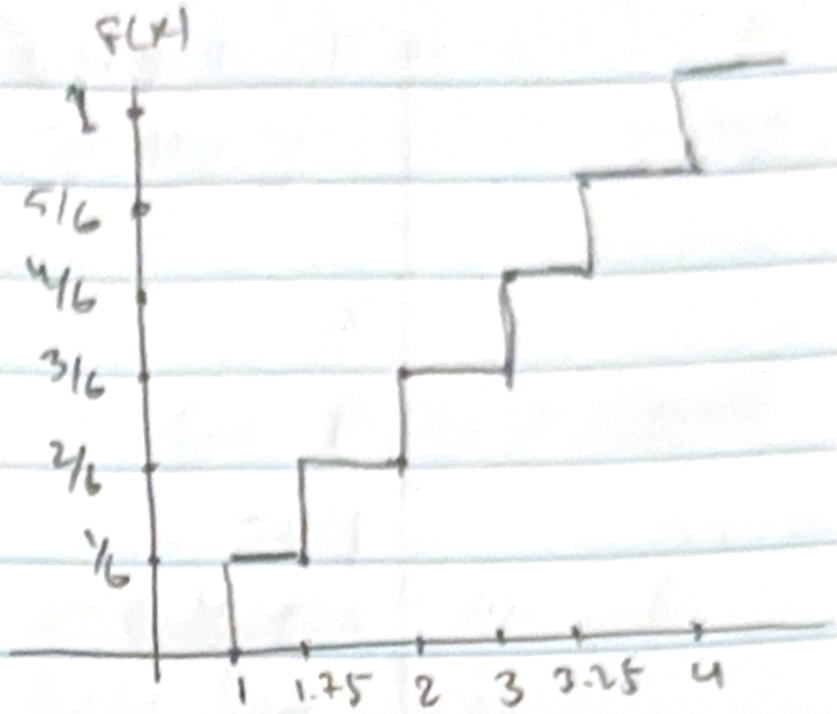
Somaya

1. $X = [1, 1.75, 2, 3, 3.25, 4]$ $n = 6$

a) Sketch the empirical cumulative distribution function

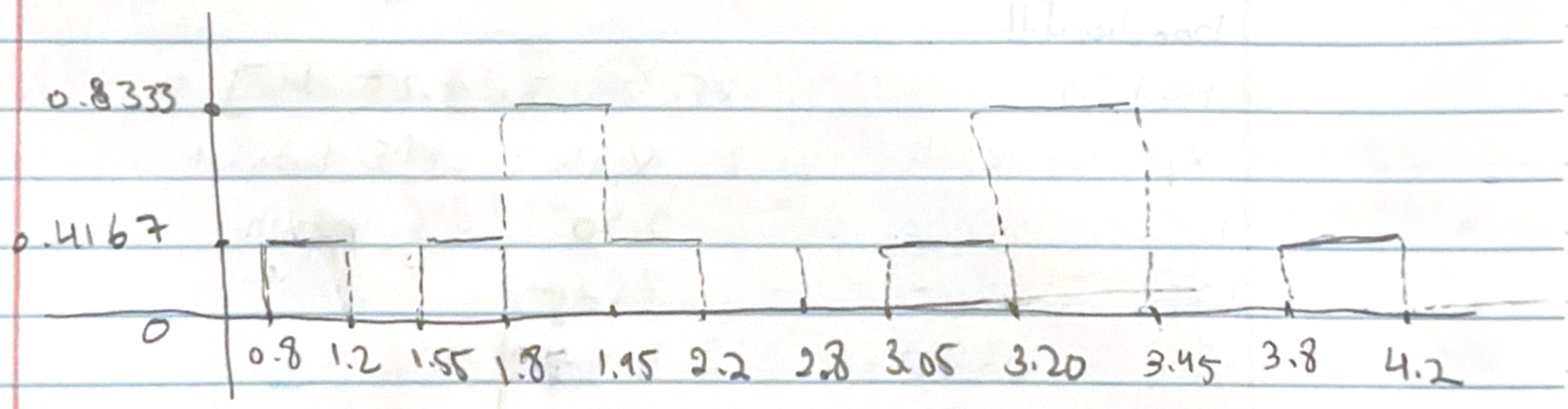
$\hat{F}(x) = \frac{\{x_i \leq x\}}{n} \cdot 1/6 = 0.1667$

x	$\hat{F}(x)$
$x < 1$	0
$1 \leq x < 1.75$	1/6
$1.75 \leq x < 2$	2/6
$2 \leq x < 3$	3/6
$3 \leq x < 3.25$	4/6
$3.25 \leq x < 4$	5/6
$x \geqslant 4$	$6/6 = 1$



b) Sketch the Uniform kernel density estimator for a bandwidth of $h = 0.2$ → kernel with bandwidth $h = 0.2$ $x_i - h$ $x_i + h$

$x = 1$	$1 - 0.2 = 0.8$	$1 + 0.2 = 1.2$	$[0.8, 1.2]$
$x = 1.75$	$1.75 - 0.2 = 1.55$	$1.75 + 0.2 = 1.95$	$[1.55, 1.95]$
$x = 2$	$2 - 0.2 = 1.80$	$2 + 0.2 = 2.20$	$[1.80, 2.20]$
$x = 3$	$3 - 0.2 = 2.80$	$3 + 0.2 = 3.20$	$[2.80, 3.20]$
$x = 3.25$	$3.75 - 0.2 = 3.05$	$3.25 + 0.2 = 3.45$	$[3.05, 3.45]$
$x = 4$	$4 - 0.2 = 3.80$	$4 + 0.2 = 4.20$	$[3.80, 4.20]$



$6 \times 1/6 = 1$ $h = 0.2$ $1 - 0.2 = 0.8$

$[0.8, 1.2]$ →

$2h = 2(0.2) = 0.4 \times \text{height}$ $\text{height} = \frac{1/6}{0.4} = \frac{1}{6(0.4)} = 0.4167$

$\hat{F}(x)$ $0.8 - 1.2 = 0.4167$

$1.2 - 1.55 = 0$

$1.55 - 1.8 = 0.4167$

$1.8 - 1.95 = 0.8333$

$1.95 - 2.2 = 0.4167$

$2.2 - 2.8 = 0$

$2.8 - 3.05 = 0.4167$

$3.05 - 3.2 = 0.8333$

$3.2 - 3.45 = 0.4167$

$3.45 - 3.8 = 0$

$3.8 - 4.2 = 0.4167$

© compute the silverman bandwidth for the uniform kernel using $h = 1.84 * SD * n ** (-0.2)$

$h = 1.84 \times SD \times n^{-0.2}$

[1, 1.75, 2, 3, 3.25, 4] $n = 6$

$SD = \frac{1+1.75+2+3+3.25+4}{6} = 15/6 = 2.5$

x_i	$x_i - 2.5$	$(x_i - 2.5)^2$
1	-1.5	2.2500
1.75	-0.75	0.5625
2	-0.5	0.2500
3	0.5	0.2500
3.25	0.75	0.5625
4	1.5	2.2500

Total = 6.125

sample SD

$SD = \sqrt{\frac{6.125}{6}} = \sqrt{1.0208333} \Rightarrow SD \approx 1.0104$

$h = 1.84 \times SD \times n^{-0.2} \Rightarrow 1.84 \times (1.1071)(6^{-0.2}) =$

$6^{-0.2} = 0.699$

$h \approx (1.84)(1.0104)(0.6988) \Rightarrow \boxed{h \approx 1.299} \Rightarrow 1.30$

Silverman's bandwidth is much larger. rectangles will be wider and we will have overlaps.

(d) Sketch the uniform kernel density estimator for Silverman's bandwidth

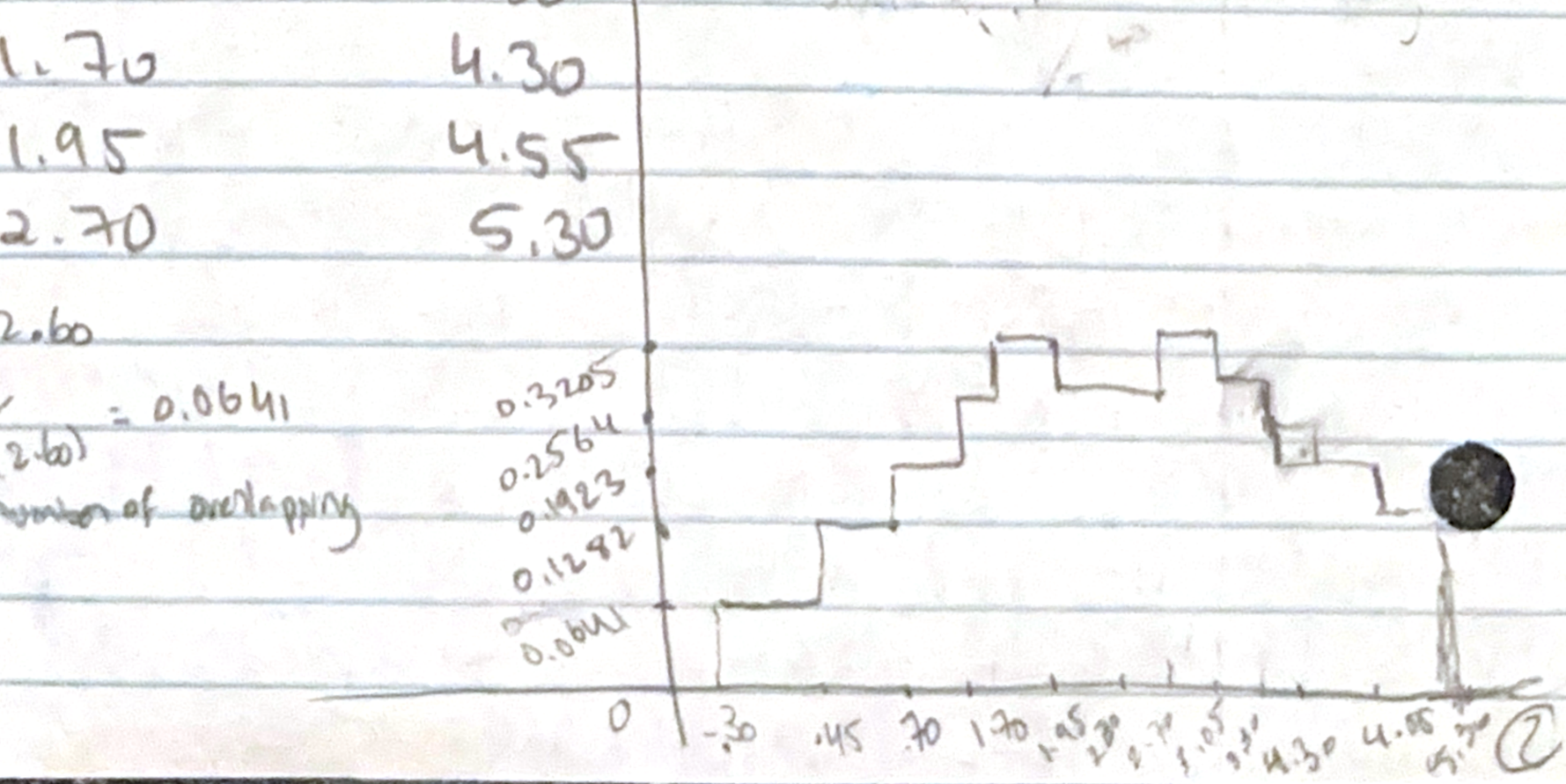
$h = 1.30$ [1, 1.75, 2, 3, 3.25, 4] $x_i - 1.30$ $x_i + 1.30$

x_i	$x_i - h$	$x_i + h$	KDE height
1	-0.30	2.30	0.0641
1.75	0.45	3.05	
2	0.70	3.30	
3	1.70	4.30	
3.25	1.95	4.55	
4	2.70	5.30	

$2h = 2(1.30) = 2.60$

$h = \frac{1}{n(2h)} = \frac{1}{6(2.60)} = 0.0641$

0.0641 * number of overlapping



(e) How are band Similar or different? Both KDEs use the Same data and uniform kernel. The KDE with h:0.2 is less Smooth and shows more detail. The Silverman bandwidth (h=1.30) produces smoother KDE because the kernels are wider and overlap more.

Problem 2:- x = [-1, -0.5, 0, 0.5, 1] drawn from Standard Normal Distribution

a) Using the uniform kernel with h=0.6, compute the realized KDE at x=0. $\hat{f}_h(0)$, directly from these five numbers.

n=5	$\hat{f}_h(0)$	x:0	h=0.6	Contain x=0	
-1		[-1.6, -0.4]	No	So at x=0 3 Kernels	
0.5		[-1.1, 0.1]	Yes	overlap	
0		[-0.6, 0.6]	Yes	height of kernel: $\frac{1}{n(2h)}$ =	
0.5		[-0.1, 1.1]	Yes		
1		[0.4, 1.6]	No	$\frac{1}{5(2\cdot0.6)} = \frac{1}{5(1.2)} = 1/6$	

Since 3 kernels overlap at 0 - = 0.1667

$\hat{F}_h(0) = 3(0.1667) = 0.5$ $\boxed{\hat{F}_h(0) = 0.5}$

(b) Now compute $E[\hat{F}_h(0)]$ at the same h=0.6.

$E[\hat{F}_h(x)] = \frac{F(x+h) - F(x-h)}{2h}$ (x=0, h=0.6) $= \frac{F(0+0.6) - F(0-0.6)}{2(0.6)} =$

$= \frac{F(0.6) - F(-0.6)}{1.2}$ => $F(0.6) = 0.7257$ $F(-0.6) = 0.2743$

$\Rightarrow E[\hat{F}_h(0)] = \frac{0.7257 - 0.2743}{2(0.6)} = \frac{0.4514}{1.2} = 0.3762$

(c) Calculate the true density at x=0 using the normal distribution.

$\mu = 0 \quad \sigma = 1$

$f(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2} = \frac{1}{\sqrt{2\pi}} e^{-0^2/2} = \frac{1}{\sqrt{2\pi}} \times 1 = \frac{1}{\sqrt{2\pi}}$

$f(0) = \frac{1}{\sqrt{2\pi}} = f(0) = \frac{1}{\sqrt{2\pi}} = 0.3989$

(d) Are b and c equal? what does it mean about bias? keep in mind, truely proving anything about bias would require us to compare $E[\hat{F}]$ to the Normal Distribution for all X; calculating this at one point is only one indication of what is going on that you can extrapolate from.

b= 0.3762 $F(0) \approx 0.3989$ That are not equal
0.3762 < 0.3982 That means at X=0, the KDE Underestimates the true density on average.
bias is at (X=0)

Bias = $E[\hat{F}_n(0)] - f(0)$ = 0.3762 − 0.3989 = −0.0227
So the true density is 0.3989. We only checked at X=0. not for all.

(e) Recompute $E[\hat{F}(0)]$ from b at a much smaller bandwidth h=0.1. what happens to the gap between $E[\hat{F}_n(0)]$ and F(0) as h shrinks

h=0 X=0 $E[\hat{F}_h(0)] = \frac{F(0+h) - F(0-h)}{2h} = \frac{F(0.1) - F(-0.1)}{0.2}$

$= \frac{0.5398 - 0.4602}{0.2} = \frac{0.0796}{0.2}$;

$E[\hat{F}_h(0)] = 0.3983$ gap = 0.3989 − 0.3762 = 0.0227

F(0) = 0.3983

as h shrinks from 0.6 to 0.1, the gap got smaller. As bandwidth gets smaller (shrinks) the gaps become smaller too. and also bias decreases

(f) Is $\hat{f}_h$ a consistent estimator? Explain your reasoning?
yes If the bandwidth h shrinks, the KDE get closer to true density
h=0.6 $\Rightarrow E[\hat{F}_h(0)] = 0.3762$ h=0.1 $E[\hat{F}_h(0)] \approx 0.3983$
for the consistency we need to have more/increase samples.
As n/samples gets larger, h approaches 0. KDE approaches true density.

(g) What are the main considerations when choosing h? Bias Variance trade off. A small bandwidth give less bias but more variance and make KDE noisy

A large bandwidth gives less variance but more bias and can oversmooth data. So h should be chosen carefully.

problem 3:- $X = [5, 7, 7, 8, 10]$ fix the bandwidth at $h = 1.5$ for both kernels below

(a) compute the uniform kernel KDE, $\hat{f}_h(x)$ at $x = 6.0$, $x = 6.5$, $x = 7.0$, $x = 8.0$, and $x = 8.5$.

height of one uniform kernel $\frac{1}{n(2h)} = \frac{1}{5(2)(1.5)} = \frac{1}{15} = 0.0667$

x	interval
5	[3.5, 6.5]
7	[5.5, 8.5]
7	[5.5, 8.5]
8	[6.5, 9.5]
10	[8.5, 11.5]

x=6	x=6.5	x=7	x=8	x=8.5
3	4	3	3	4

$\hat{f}_h(6.0) = 3(0.0667) = 0.2000$

$\hat{f}_h(6.5) = 4(0.0667) = 0.2667$

$\hat{f}_h(7) = 3(0.0667) = 0.2000$

$\hat{f}_h(8) = 3(0.0667) = 0.2000$

$\hat{f}_h(8.5) = 4(0.0667) = 0.2667$

(b) compute the Gaussian kernel KDE at the same five values of x, using the same $h = 1.5$

$$\hat{f}_h(x) = \frac{1}{nh} \sum_{i=1}^{n} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x - x_i}{h}\right)^2} \qquad nh = 5(1.5) = 7.5$$

$$\hat{f}_h(6) = \frac{1}{7.5}\left[\phi\frac{6-5}{1.5} + \phi\frac{6-7}{1.5} + \phi\frac{6-7}{1.5} + \phi\frac{6-8}{1.5} + \phi\frac{6-10}{1.5}\right]$$

$$\hat{f}_h(6) = [\phi 0.6667 + 0.31945 + 0.31945 + 0.16401 + 0.01140]$$

$$\hat{f}_h(6) = \left(\frac{1.13376}{7.5}\right) \cdot \hat{f}_h(6) \approx 0.1512$$

$$\hat{f}_h(6.5) = \frac{1}{7.5}\left[\phi(1) + \phi(-0.3333) + \phi(-0.3333) + \phi(-1) + \phi(-2.3333)\right]$$

$$= \frac{1}{7.5}\left[0.24197 + 0.37738 + 0.37738 + 0.24197 + 0.02622\right] =$$

$$\frac{1.26492}{7.5} = \hat{f}_h(6.5) = 0.1687$$

$\hat{F}_h(8) = \frac{1}{7.5}[\phi(2) + \phi(0.6667) + \phi(0.6667) + \phi(0) + \phi(-1.3333)]$

$= \frac{1}{7.5}[0.05399 + 0.31945 + 0.31945 + 0.39894 + 0.16401]$

$= \frac{1.25584}{7.5} = \boxed{\hat{f}_h(8) = 0.1674}$

$\hat{F}_h(7) = \frac{1}{7.5}[\phi(1.3333) + \phi(0) + \phi(0) + \phi(0) + \phi(-0.6667) + \phi(2)]$

$= \frac{1}{7.5}[0.16401 + 0.39894 + 0.39894 + 0.31945 + 0.05399]$

$= \frac{1.33533}{7.5} = \boxed{0.1780}$

$F_h(8.5) = \frac{1}{7.5}[\phi(2.3333) + \phi(1) + \phi(0.3333) + \phi(1)]$

$\frac{1}{7.5}[0.02622 + 0.24197 + 0.24197 + 0.37738 + 0.24197]$

$= \frac{1.12951}{7.5} = \hat{f}_h(8.5) = 0.1506$

(c) Your part a values should jump abruptly between some of these five points, while your part b values should change smoothly. Identify where the biggest jump in (a) happens, and explain in terms of "two" kernel shapes - not just one is "smoother" - why that jump exists in the uniform case but not the Gaussian case?

KDE values from (a)

x	6	6.5	7	8	8.5
$\hat{f}_h(x)$	0.2000	0.2667	0.2000	0.2000	0.2667

$6 \to 6.5$: $0.2000 \to 0.2667$

$6.5 \to 7.0$: $0.2667 \to 0.2000$

$8.0 \to 8.5$: $0.2000 \to 0.2667$

So each has a change of 0.0667

Gaussian KDE changes smoothely

The Uniform Kernel is rectangle and has positive height. it suddenly goes to zero. So, it creates jumps.

Gaussian is bell shaped and has no sharp cutoff. As x moves away from data points, it gradually get smaller rather than becoming zero.

d. Why might you prefer one kernel over the others in practice?
In Practice, we may prefer Gaussian kernel because it doesn't jump abruptly. It produces a smooth density estimate. It bell shaped changes gradually. The uniform kernel is simple to compute but it jumps.

problem 4

c. for what values of ejection_fraction is the original plot reliable? For which values is there noticable variation in your plots of resampled data?

The original KDE is most reliable for ejection_fraction values from about 20 to 30 and 40-50, where resampled curves are close together. There is noticible variation around 35-40 and 55-65 where resampled KDE curves are more spread out.